Do Now · retrieval

Final mixed-paper readiness

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Difference between MAR and MDR?

Q2

How to encode a negative integer in two's complement?

Q3

What does $O(n^2)$ describe?

Q4

Full adder A=1 B=1 carry-in=1. State sum and carry.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Identify — Retrieve knowledge across both components

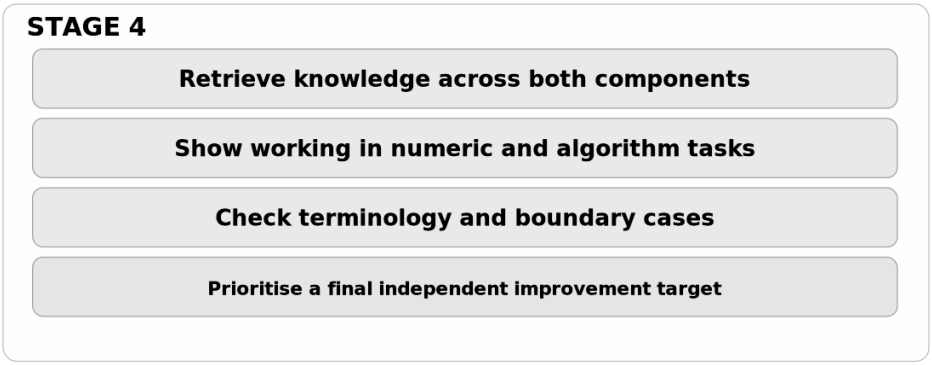
Core idea 2

Model — Show working in numeric and algorithm tasks

Core idea 3

Apply — Check terminology and boundary cases

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Core idea 4

Evaluate — Prioritise a final independent improvement target

Vocabulary · use the correct technical terms

readiness — Ability to apply without prompts.

synoptic — Knowledge across multiple topics.

method evidence — Reasoning supporting an answer.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Normalise positive $0.0101_2 \times 2^3$.

Follow the worked method

1. Explain the method — Shift mantissa one position left and reduce exponent by one.

Source model · final illustrated step

IDO · MODEL

Shift mantissa one position left and reduce exponent by one.

Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Explain the method
Each midpoint comparison reduces the interval; return index4.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Q1 [8]: In a simplified sequential fetch, PC=12 and memory address 12 stores instruction ADD 7. State MAR, MDR, CIR and next PC (assume no branch or interrupt). [4] Convert −5 to 8-bit two's complement. [4]

Question 2

Q2 [8]: Trace binary search for 11 in [1,3,5,7,9,11,13]. [4] Explain base case and recursion progress. [4]

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Q3 [9]: Evaluate cloud storage versus USB-only storage for an inclusive school revision service; weigh access, reliability, security. [9]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Publishing every lesson guarantees each student is ready for assessment.

Use key vocabulary from memory

1. Explain readiness in your own words.

2. Explain synoptic in your own words.

3. Explain method evidence in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.